RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 4: INFORMATION RETRIEVAL USING NLTK

QUESTION:
An e-commerce company receives thousands of customer reviews daily. Develop an information retrieval system using the Amazon Fine Food Reviews dataset to preprocess customer reviews, convert them into TF-IDF feature vectors, and retrieve the five most relevant reviews for a user query using cosine similarity.

AIM:
To implement an information retrieval system using NLTK by cleaning and preprocessing customer reviews, converting text into TF-IDF feature vectors, calculating cosine similarity, and retrieving the five most relevant reviews for a given search query.

ALGORITHM
1. Import the required Python libraries, including Pandas, NLTK, Scikit-learn, and cosine similarity utilities.
2. Load the Amazon Fine Food Reviews dataset into a Pandas DataFrame and retain the first 10,000 reviews.
3. Remove missing and duplicate reviews and select the review text column.
4. Preprocess the reviews by converting text to lowercase, removing punctuation, numbers, and special characters, tokenizing words, removing English stop words, and joining the cleaned tokens.
5. Store the cleaned reviews in a new column named Cleaned_Text and display five sample reviews.
6. Convert the cleaned reviews into TF-IDF feature vectors and display the matrix shape and vocabulary size.
7. Create a retrieval function that preprocesses a user query, converts it into a TF-IDF vector, and calculates cosine similarity with all review vectors.
8. Retrieve and display the top five most relevant reviews along with their similarity scores and original review text.
9. Test the retrieval system using different queries and compare the relevance of the retrieved reviews.

PROGRAM & OUTPUT:

```
# Name: Shyam M (241801267)
# Section 1: Import Required Libraries

import pandas as pd
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import wordpunct_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

nltk.download("stopwords")
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
# Name: Shyam M (241801267)
# Section 2: Load and Clean Dataset

df = pd.read_csv(
    "/content/Reviews.csv",
    on_bad_lines="skip",
    engine="python"
)

df = df.dropna(subset=["Text"])
df = df.drop_duplicates(subset=["Text"])
df = df.head(10000).copy()

print("Dataset shape:", df.shape)
print(df[["Text"]].head())
```

```
Dataset shape: (10000, 10)
                                                Text
0  I have bought several of the Vitality canned d...
1  Product arrived labeled as Jumbo Salted Peanut...
2  This is a confection that has been around a fe...
3  If you are looking for the secret ingredient i...
4  Great taffy at a great price.  There was a wid...
```

```
# Name: Shyam M (241801267)
# Section 3: Text Preprocessing

stop_words = set(stopwords.words("english"))

def preprocess(text):
    text = str(text).lower()
    text = re.sub(r"[^a-z\s]", " ", text)
    tokens = wordpunct_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    return " ".join(tokens)

df["Cleaned_Text"] = df["Text"].apply(preprocess)

print("\nOriginal and cleaned reviews:")
print(df[["Text", "Cleaned_Text"]].head(5))
```

```
Original and cleaned reviews:
                                                Text  \
0  I have bought several of the Vitality canned d...
1  Product arrived labeled as Jumbo Salted Peanut...
2  This is a confection that has been around a fe...
3  If you are looking for the secret ingredient i...
4  Great taffy at a great price.  There was a wid...

                                        Cleaned_Text
0  bought several vitality canned dog food produc...
1  product arrived labeled jumbo salted peanuts p...
2  confection around centuries light pillowy citr...
3  looking secret ingredient robitussin believe f...
4  great taffy great price wide assortment yummy ...
```

```
# Name: Shyam M (241801267)
# Section 4: Convert Reviews into TF-IDF Vectors

vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(df["Cleaned_Text"])

print("TF-IDF matrix shape:", tfidf_matrix.shape)
print("Vocabulary size:", len(vectorizer.vocabulary_))
```

```
TF-IDF matrix shape: (10000, 18523)
Vocabulary size: 18523
```

```
# Name: Shyam M (241801267)
# Section 5: Information Retrieval Function

def retrieve_reviews(query, top_n=5):
    cleaned_query = preprocess(query)
    query_vector = vectorizer.transform([cleaned_query])

    similarity = cosine_similarity(
        query_vector, tfidf_matrix
    ).flatten()

    top_indices = similarity.argsort()[-top_n:][::-1]

    results = df.iloc[top_indices][["Text", "Cleaned_Text"]].copy()
    results["Similarity"] = similarity[top_indices]

    return results
```

```
# Name: Shyam M (241801267)
# Section 6: Test Sample Queries

queries = [
    "great product with fast shipping",
    "disappointed",
    "excellent taste",
    "poor packaging"
]

for query in queries:
    print("\nSearch Query:", query)
    results = retrieve_reviews(query)

    for rank, (_, row) in enumerate(results.iterrows(), start=1):
        print("\nRank:", rank)
        print("Similarity:", round(row["Similarity"], 4))
        print("Review:", row["Text"][:500])
```

```
Search Query: great product with fast shipping

Rank: 1
Similarity: 0.5103
Review: Enjoyed the product and they also provided very fast shipping.  I'm about out and need to order more.

Rank: 2
Similarity: 0.4925
Review: The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.<br />I recommend this prc

Rank: 3
Similarity: 0.4001
Review: These are very good, they were a great price and fast-free shipping when you purchase enough food.  The only t

Rank: 4
Similarity: 0.3996
Review: The energy drink is a great product. The shipping price is crazy tho. I paid more for the shipping then both p

Rank: 5
Similarity: 0.3936
Review: Was exactly what I was looking for. Super fast shipping! A bit on the spicy side,but great taste!!!!<br />Woul

Search Query: disappointed

Rank: 1
Similarity: 0.4846
Review: I am a bit disappointed.  The flavor was not what I wanted or expected.

Rank: 2
Similarity: 0.3582
Review: The product is very good. Way too expensive and at almost 5.00 a box to get pancakes or biscuits I was disappc

Rank: 3
Similarity: 0.3216
Review: I was disappointed in this product because I thought it would be bigger.<br />Also, it did not come with enoug

Rank: 4
Similarity: 0.3195
Review: Disappointed.  The big boxes had a very different flavor than the smaller boxes.  I am not sure why but not gc

Rank: 5
Similarity: 0.3117
Review: Just plain nasty!!! This item tasted like cardboard, very watered down and no flavor to it at all.  Very disap

Search Query: excellent taste

Rank: 1
Similarity: 0.502
Review: It is very very good sweets<br />Amazon always is excellent<br />Amazon always is excellent<br />Amazon always

Rank: 2
Similarity: 0.4605
Review: The Popchips were excellent - had excellent flavor and low in calories.  When you eat these - don't need anoth

Rank: 3
Similarity: 0.4054
Review: Excellent product. Exactly what they said it would be. Superior chamomile from what I have ordered before from

Rank: 4
Similarity: 0.3856
Review: We recieved the product on time and in excellent shape. It tastes good and meets our expectations. Thanks for

Rank: 5
Similarity: 0.384
Review: Title says it all, a Excellent quality product and Excellent service at a fair price. Product came ahead of ti

Search Query: poor packaging

Rank: 1
Similarity: 0.3177
Review: I was delighted to find this product online, because I cannot always find it in the stores.  However, each pac

Rank: 2
Similarity: 0.3085
Review: This review is for the packaging and not the actual mix, since the mix is a very subjective thing and my taste

Rank: 3
```

```
Review: This review is for the packaging and not the actual mix, since the mix is a very subjective thing and my taste

Rank: 3
Similarity: 0.3083
Review: The soup is rich and tasty, but DON'T buy it from Amazon.  The packaging is so poor that the cans arive dentec

Rank: 4
Similarity: 0.292
Review: Didnt really enjoy these after wanting to try something other than aunt jemima's from Dominicks. The pancakes

Rank: 5
Similarity: 0.2811
Review: was really dissappointed in product... too salty and poor flavor....Convenience is what makes product worth pu
```

```
# Name: Shyam M (241801267)
# Section 7: Search Using a Custom Query

query = input("Enter your search query: ")
print("\nTop 5 Relevant Reviews:")
print(retrieve_reviews(query).to_string(index=False))
```

```
... ir your search query: great product with fast shipping

    5 Relevant Reviews:

                                                                Enjoyed the product and they also provided very fast sh
                                                        The tea is good and fresh. We enjoy it. The shipping is fast and cos
    ie are very good, they were a great price and fast-free shipping when you purchase enough food.  The only thing is that
                                The energy drink is a great product. The shipping price is crazy tho. I paid more for the shipping th
                                    Was exactly what I was looking for. Super fast shipping! A bit on the spicy side,but great taste
```

RESULT:

Thus, an information retrieval system was successfully implemented using NLTK and TF-IDF vectorization on the Amazon Fine Food Reviews dataset. Customer reviews were cleaned and preprocessed, converted into TF-IDF feature vectors, and compared using cosine similarity. The system retrieved the top five most relevant reviews for different search queries and displayed their similarity scores, helping users identify relevant customer feedback efficiently.